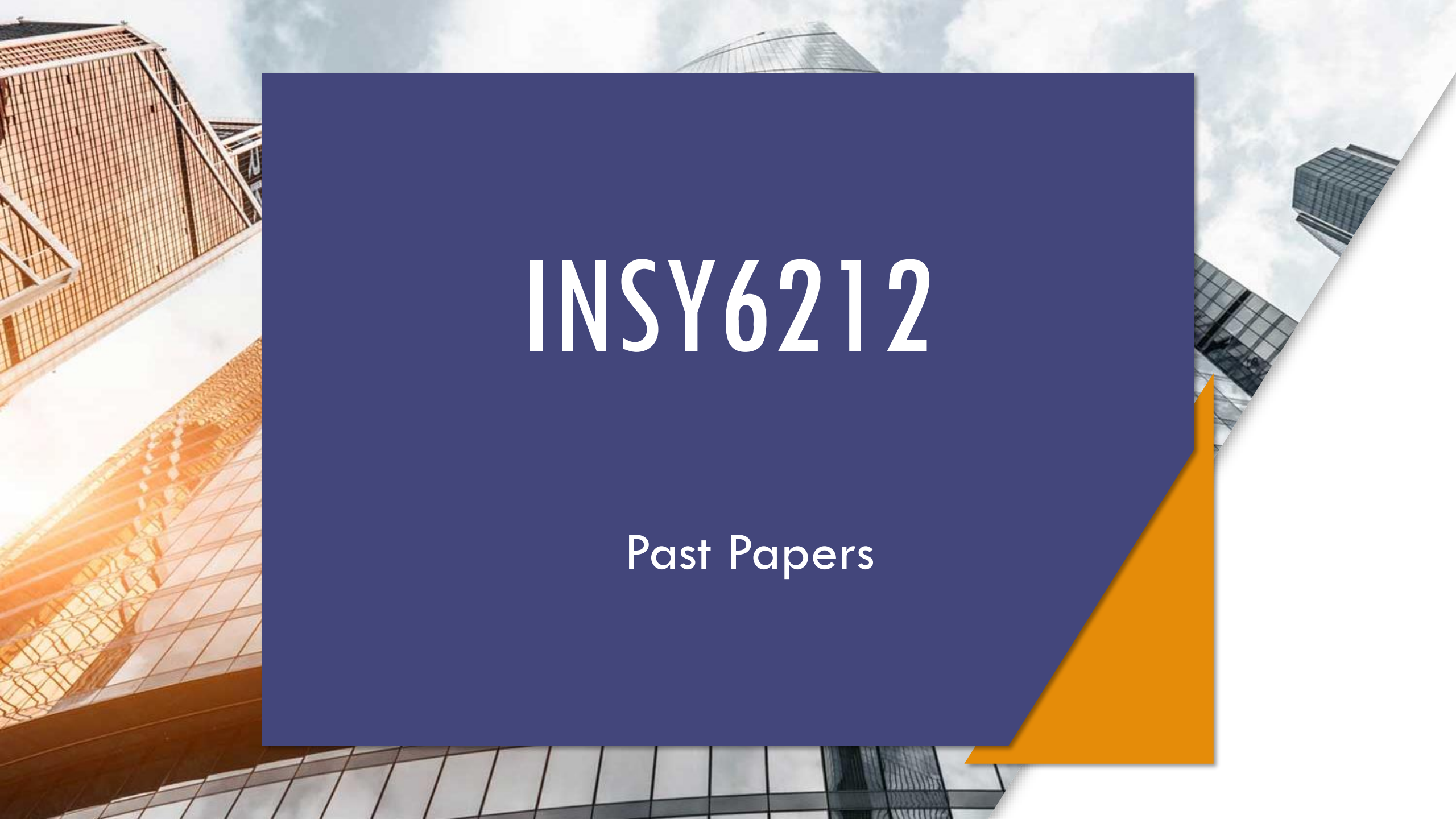


INSY6212

Past Papers

Paper 1

Question 1		(Marks: 15)
Consider a project you were once involved in or are currently involved in e.g., your WIL project, project from any other module, etc.		
Q.1.1	Briefly explain the objective of your project then describe four actions you will need to take when you need to start closing the project.	(10)
Q.1.2	Discuss is the importance of documenting lessons learned, as well as archiving project documentation after your project has concluded.	(5)

1.1. LU1 and LU4 (chapter 4 and 9)

- Explain the project objective for 2 marks
- Describe the project closing actions – 2 marks for each action explained
- Collecting and making final payments
- Recognising and evaluating staff
- Conducting a post-project evaluation
- Documenting lessons learnt
- Organising and archiving project documents

1.2. LU4 (chapter 9 – page 316 and 317)

- Discuss the importance of documenting lessons learnt – 2 marks
- Discuss the importance of archiving project documentation – 2 marks
- Link the discussions back to the identified project – 1 marks

Notes:

Importance of documenting lessons include:

- **Capitalising on the knowledge and experience gained on the project to improve future project performance.**
- **To create a base of knowledge that is easily retrievable by category or subject or keyword.**
- **etc**

importance of archiving project documentation include:

- **To make documentation readily retrievable for future use**
- **To create a database of common elements that can be used for project and trend analysis.**
- **Etc.**

Question 2 (Marks: 30)

Answer the questions fully.

Q.2.1	Explain what a functional organisation structure is and clearly highlight at least two advantages and two disadvantages of this structure.	(5)
Q.2.2	Discuss the role of collaborative communication tools in a project. Provide two examples of such tools and how they are utilised in a project	(5)
Q.2.3	Discuss the importance of a well-defined project evaluation and selection process in an organisation. Clearly articulate the impact of not having such a process using relevant examples.	(5)
Q.2.4	Explain the importance of controlling changes made to project schedule. How can you achieve effective schedule control?	(5)
Q.2.5	Provide five guidelines for ensuring the usefulness and value of project reports to their recipients. For each guideline, justify its importance.	(10)

M.2.1	*Ref: LU9* • *Award one mark for the discussion on what a functional organisational is.* **Individuals are organised in groups that perform the same function or have the same expertise or skills.** • *Award one mark for each advantage to a maximum of two marks.* • **No duplication of activities** • **Functional Excellence** • *Award one mark for each disadvantage to a maximum of two marks.* • **Insularity** • **Slow response time.**	**(5)**

2.2. LU9

- Collaborative Tools allow collaborative workspaces between members of the project team and subcontractors and customer to communicate and collaborate.

2.3. LU2

- A well-defined and well-understood project evaluation and selection process increase the chances of making the best project decision

2.4. Chapter 5

- So that the processes and procedures that define how changes to schedule will be documented, approved, and communicated can be managed.
- To achieve effective schedule control, one must:

1. Analyse schedule to determine areas which may need corrective action.
2. Decide what specific corrective actions should be taken.
3. Revise the plan to incorporate chosen corrective action.
4. Recalculate the schedule to evaluate the effects of planned corrective actions.

2.5. **Not covered in 2025 syllabus**

Question 4 **(Marks: 5)**

For the project you identified in **Q.1**, state any five risks that could jeopardise the success of the project.

Question 5 **(Marks: 30)**

Think of an **IT project** suitable for Agile project management and answer the following questions:

Q.5.1	Discuss how an agile release plan differs from a traditional work breakdown structure. And justify which one would be more beneficial to use in a project.	(5)
Q.5.2	Discuss five ways in which scope creep is or can be caused in an agile project.	(5)
Q.5.3	In a tabular format, describe five (5) differences between the traditional project management approach and the agile project management approach and provide an example of each.	(15)
Q.5.4	When working with Microsoft project, you must remember that the project is a relational database. What does this mean? And why is it important to note?	(5)

4. LU8 Chapter 8

- Weather
- Lack of funding
- Lack of resources – skilled people, materials, facilities, etc
- Changes to govt regulations
- Protestors
- Delays in approvals

M.5.1	*Ref: Articles* • *Award two marks for defining what an agile release plan is.* It maps features and requirements such as epics and user stories out to give a list of features to be built over time. It is a high level WBS and project schedule in one. • *Award one mark for each difference identified to a maximum of three marks.* • The focus is on functionality delivered to production in a collaborative manner rather than interim deliverables aligned with functional role responsibilities. • It allows room to leave doing the job to the professionals on the team rather than imprinting the job description in every project plan. • It allows the acceptance that planning should be refined over time, from high level to detailed rather than trying to plan everything at project initiation.	(5)

5.2.

- Define what scope creep is and how it is caused in agile project management

5.3.

Characteristic	Traditional	Agile
Requirements	Clear initial requirements; low change rate (1/2)	Creative, innovative; requirements unclear (1/2)
Users	Not involved (1/2)	Close and frequent collaboration (1/2)
Documentation	Formal documentation required (1/2)	Tacit knowledge(1/2)
Project Size	Bigger projects (1/2)	Smaller projects (1/2)
Organisational support	Use existing processes; bigger organisations (1/2)	Prepared to embrace agile approach (1/2)
Team members	Not accentuated; fluctuation expected; distributed team (1/2)	Collocated team; smaller team (1/2)
System criticality	System failure consequences serious (1/2)	Less critical systems (1/2)
Project Plan	Linear (1)	Complex; iterative (1)

M.5.4	*Ref: Articles* • *Award one mark for explaining why Microsoft project is a relational database.* It frees up the mind of those who are using it and allows the tool to be used with agile, waterfall or any mix of these methodologies. • *Award three marks for discussing the importance of noting that Microsoft project is a relational database.* You can apply this mindset and approach the building of database elements into your schedule in a way that helps you manage in an agile view and switch back to waterfall (2), including elements necessary for filtering, grouping, sorting, and reporting. (1)	(5)

Question 6 (Marks: 20)

Read the scenario below and answer the question that follow:

You have been contracted to work on an e-hailing platform for a new start-up that seeks to challenge the industry leaders. They need a rockstar development team that is open to the ever-changing technology landscape, as they would like to always make sure that their platform is running on the most up-to-date technology. They have a base plan of requirements and are still liaising with their internal analysis team to flesh out some more requirements for the platform. They know it will take time to build the full-fledged platform but would like to roll out a minimal happy product to consumers to start off with and to also get some real-time customer feedback apart from the internal requirements still to be designed and developed.

Explain, with the aid of relevant examples, how each of the following agile methodologies embrace and manage changing scope throughout the life of the project:

- Scrum
- Extreme Programming
- Kanban

- Scrum is a subset of Agile and one of the most popular process frameworks for implementing agile. It is an iterative development model used to manage complex software and product development.

- Extreme Programming is a set of technical practices for software developers that turn the dial up to 11 on practices that generally considered to be good things to do. This becomes a collaborative exercise, one in which programmers can learn from each other while assisting in the creation of high-quality efficient codes.

- Kanban is a visual framework used to implement agile and shows what to produce, when to produce it, and how much to produce. It encourages small incremental changes to your current system and does not require set up or procedure.

Paper 2

Question 1		(Marks: 10)
As a project manager, you need to understand how project communication and documentation work. You have two project members who can never seem to see eye to eye, and this causes a lot of conflict, especially when deliverables are due. You have your suspicions but want to cover your bases to make sure.		
Q.1.1	Identify two barriers to effective listening and explain how exactly they affect effective listening.	(5)
Q.1.2	Discuss the purpose of a status review meeting and identify three agenda items that might be discussed at such a meeting.	(5)

1.1. LU9

- Pretending to listen
- Distractions
- Bias and closed-mindedness
- Impatience
- Jumping to conclusions

1 mark for each barrier and 2 marks for explaining how the barriers affect listening

1.2. Not covered in 2025 Syllabus

Question 2		**(Marks: 30)**
Answer the questions fully.		
Q.2.1	Discuss what an activity duration is and how it is determined.	(5)
Q.2.2	Explain the role of the scrum master as compared to the role of a project manager.	(5)
Q.2.3	Discuss the changes that commonly become necessary during information system development projects.	(10)
Q.2.4	Discuss the different types of contracts and give an example of a project where each kind of contract would be used.	(10)

2.1. LU6 Chapter 6

Award two marks for discussing what an activity duration is

The total elapsed time for an activity

Award three marks for explaining how it is determined.

Once the types and quantities of resources are estimated for each activity, estimates can be made for how long it will take to perform each activity (2). It must be the total time for the work to be done plus any associated waiting time (1)

2.2. LU articles

Award two marks for defining what a scrum master is.

Award two marks for defining the role of a project manager.

Award one mark for differentiating the two roles.

2.3.

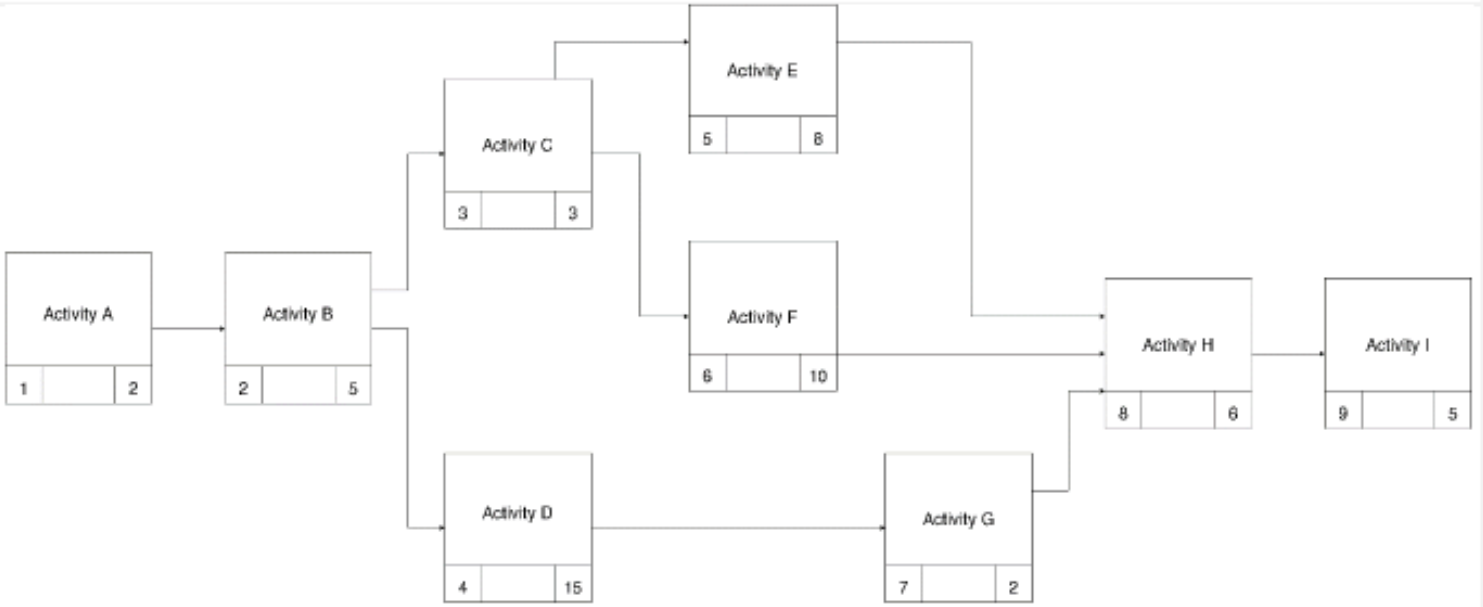
Award one mark for each change to a maximum of five marks

- Changes to the interface
- Changes to reports
- Changers to online queries
- Changes to software processing routines
- Changes to storage capacities

Award one mark for each change description to a maximum of five marks

M.2.4	*Ref: LU3* *Award two marks for the description of each type of contract to a maximum of four marks* • Fixed price contract • Cost reimbursement contract *Award three marks for an example of a project where a fixed-price contract can be used.* *Award three marks for an example of a project where a cost-reimbursement contract can be used*	(10)

Question 3 (Marks: 20)



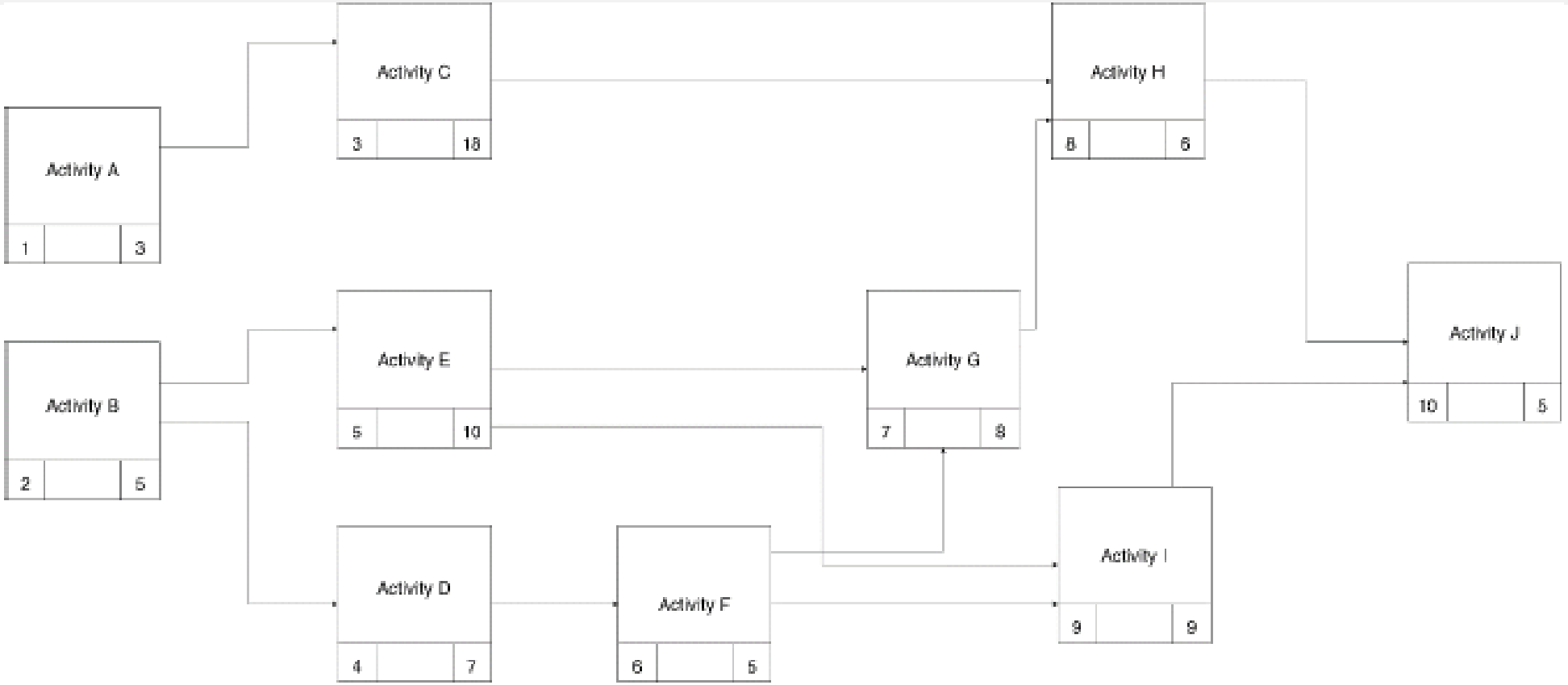
Q.3.1 Consider the network diagram above and Calculate the Earliest Start, Earliest Finish, Latest Start, and Latest Finish times, along with the total slack for each activity. (20)

Complete in a table similar to the below.

Activity	*Estim. Dur.*	*Earliest*		*Latest*		*Total Slack*
		Start	*Finish*	*Start*	*Finish*	

Question 4 (Marks: 10)

Consider the network diagram below and answer the questions that follow:



Q.4.1 Draw up a list of activities and Immediate Predecessors for the network diagram. (10)

M.4.1 *Ref: LU1* . (10)

Award eight marks based on the table below.

Activity	Immediate Predecessors	Estimated duration
A	-	3
B	-	5
C	A (1)	18
D	B (1)	7
E	B (1)	10
F	D (1)	5
G	E, F (1)	8
H	C, G (1)	6
I	E, F (1)	9
J	H, I (1)	5

Award two marks for showing the activity list in a professional, tabular format

Q.5.3	Research the following five agile project management tools, discuss their role and purpose in an agile project and give examples of how they can be used in your identified project above. • Microsoft Project • ClickUp • Jira • DailyScrum • Notion	(10)
Q.5.4	Discuss what Milestones are and the importance of adding deadlines to Milestones in an agile project.	(10)

Microsoft Project

- Microsoft Project is a project management software product developed and sold by Microsoft. It is designed to assist a project manager in developing a schedule, assigning resources to tasks, tracking progress, managing the budget, and analysing workloads.

ClickUp

- designed for engineering and product teams of any size to plan, organise, and collaborate on work using tasks, docs, chat, goals, whiteboards, and more. It's a powerful Agile project management solution known for its intuitive UI and rich set of customisable features to increase efficiency across the board—whether you're managing workflows, bugs, releases, or wikis.

Jira

- Tracking tool that is used for Agile testing as well as project management. It is developed for bug tracking, issue tracking and project management for software and mobile development processes.

DailyScrum

- system built to plan, track and manage Agile projects. Some key features like release planning and project forecasting, project control, sprint planning, and system integration

Notion

- Communicate, making it easy to add team members. The notion is all you need for your notes, tasks, and wikis to discuss projects, share files, and record ideas.

M.5.4	*Ref: LU Articles* *Award two marks for the definition of what a milestone is.* A tool that marks a specific point along a project timeline *Award eight marks for discussion on the importance of adding deadlines to milestones.* • The schedule is given structure, although it may be stuck in time or constrained. (2) • It keeps your activities from moving beyond sprint windows. • It allows your teams to report progress at regular intervals (2) • Provides the ability to have a project alert if any remaining work slips beyond an overall sprint time (2) • Indicates to the project manager that something needs to be done to bring the schedule in line (2)	(10)

Paper 3

Question 1 **(Marks: 15)**

You have been contracted to put together a team to develop a pothole detection mobile app for the Department of Road Works. It is your responsibility to source, interview, assign responsibilities to the team, and draw up a proposal for the budget of the mobile application.

Q.1.1	Discuss three key responsibilities of a project manager and how they apply to the case above.	(5)
Q.1.2	A project manager should have negotiation skills. Discuss what the goal of negotiation is and how it is helpful as a skill for the project manager. Provide a scenario based on the project above highlighting the use of negotiation skills.	(10)

M.1.1	*Ref: LU9* • *Award one mark for each responsibility discussed to a maximum of three marks.* • *Award two marks for applying these responsibilities to the case study.* Notes: • Planning • Organising • Controlling	(5)

M.1.2	*Ref: LU9 p.345* Answers may vary. Marker's use marking guide provided. • *Award two marks for the discussion on what the goal of negotiation is.* • *Award up to three marks for discussing how it is a helpful skill to have for the project manager.* • *Award a total of five marks for student's scenario based on the project as follows:* ○ *0 – No scenario provided.* ○ *1-2 marks: Inadequate scenario provided, linked to project, unclear explanation of negotiation skills in project.* ○ *3 marks: Adequate scenario provided, linked to project, explanation of negotiation provided, could be improved to make it clear and concise.* ○ *4-5 marks: Excellent scenario provided, linked to project, clear and concise explanation of negotiation provided.*	(10)

Notes:

- For two people or parties to arrive at a mutually acceptable agreement on an issue.
- It is a form of problem solving, that results in a win-win outcome for both parties.

Question 2		**(Marks: 30)**
Answer the questions fully		
Q.2.1	Discuss five characteristics of effective teams.	(5)
Q.2.2	Discuss any two dimensions of diversity when it comes to valuing team diversity on a project. Provide examples to illustrate how these impact IT projects.	(10)
Q.2.3	Provide a brief background of the project in which you were involved or were the project manager, clearly stating whether it was a success or failure. List three lessons learned from a project.	(5)
Q.2.4	Define the following and show how each is calculated: • FCAC • CPI	(10)

M.2.1	*Ref: LU9* • *Award one mark per characteristic discussed to a maximum of five marks.* • *Award ½ mark for merely stating the characteristic.* *Notes:* • *Clear understanding of the project objective.* • *Results orientation* • *High degree of cooperation and collaboration.* • *High level of trust* • *Clear expectations of roles and responsibilities.*	(5)

M.2.2	*Ref: LU9* • *Award one mark per dimension to a maximum of two marks.* • *Award two marks for discussion to a maximum of four marks.*	(10)
	• *Award two marks for each adequate example to a maximum of four marks.* • *Award a total of ten marks.* <u>Dimensions include:</u> • Age or generation • Job status • Religious Affiliation • *Ethnicity* • *Appearance* • *Health* • *etc*	

M.2.3	***Ref: LU9*** • *Award two marks for brief background of the project and what went wrong or what went well.* • *Award three marks for documenting the lessons learned during the project* • *Award a total of five marks.*	**(5)**

M.2.4	*Ref: LU7* • *Award two marks for each definition to a maximum of four marks.* • *Award three marks for each calculation to a maximum of six marks.* • *Award a total of ten marks.* **Notes:** • **FCAC (Forecasted Cost at Completion)** • **CPI (Cost performance Index)**	**(10)**
	Calculations: • **FCAC** ○ **Forecasted cost at completion(1) = Total Budgeted cost(1)/Cost performance index(1)** • **CPI** ○ **Cost performance index (1) = Cumulative earned value (1)/Cumulative actual cost (1)**	

Question 4 **(Marks: 5)**

Consider the project scenario in **Q.1**:

Draw up a list of five risks that could jeopardise this project under the following categories:

- 2 Technical risks
- 2 Human Resources risks
- 1 External risk

Ref: LU8

- *Award one mark for each risk identified under Technical risks to a maximum of two marks.*
- *Award one mark for each risk identified under Human Resources risks to a maximum of two marks.*
- *Award one mark for each risk identified under* ***External*** *risks to a maximum of one mark.*
- *Award a total of five marks.*

Question 6 **(Marks: 20)**

Read the scenario Below and answer the questions that follow:

You have been contracted to work on an e-hailing platform for a new start-up that seeks to challenge the industry leaders. They need a rockstar development team that is open to the ever-changing technology landscape, as they would like to always make sure that their platform is running on the most up-to-date technology. They have a base plan of requirements and are still liaising with their internal analysis team to flesh out some more requirements for the platform. They know it will take time to build the full-fledged platform but would like to roll out a minimal happy product to consumers to start off with and to also get some real-time customer feedback apart from the internal requirements still to be designed and developed.

The Agile Manifesto lists 12 principles to guide teams on how to execute with agility:

With regards to the agile principles, discuss how you would use six of the twelve principles to ensure the successful completion of the above project.

Memorandum: Question 6 **(Marks: 20)**

Ref: LU1 Articles

Award one mark per principle identified to a maximum of six marks.

- Satisfy the customer.
- Welcome changing requirements.
- Deliver working software frequently.
- Face-to-face conversation.
- Simplicity
- Reflect and adjustment.

- *Award one mark for discussing how each principle works to a maximum of six marks.*
- *Award one mark for discussing how each principle can be used to successfully complete the project to a maximum of six marks.*
- *Award two marks for a cohesive discussion that links back to the case.*
- *Award a total of twenty marks.*

Paper 4: LU8 and LU7

Question 2		(Marks: 15)
Q.2.1	Project Management is a skill that needs to constantly be developed. Identify one skill a Project Manager requires then: • Explain the importance of the skill in a project. • List three ways to develop the skill.	(5)
Q.2.2	Brainstorming is a technique used in problem solving. Discuss any four tips for conducting a successful brainstorming session.	(8)
Q.2.3	Briefly explain how a matrix organisational structure provides for career development.	(2)

Question 4 **(Marks: 20)**

Below is a table of the total budgeted costs in thousands, for each task in a project. Answer **Q.4.1** that follows.

Activity ID	TBC	Month										
		1	2	3	4	5	6	7	8	9	10	11
A	30	10	15	5								
B	100		10	10	10	10	5	5	10	10	10	20
C	80			10	15	15	20	20				
D	130		15	15	10	10	25	25	30			
E	10	10										
F	40									20	20	
G	20										10	10
H	10											10
Total	420											
Cumulative												

Q.4.1	Using the table above, calculate the cumulative budgeted costs for the following months: a. Month 1	(6)
	b. Month 3 c. Month 10	
Q.4.2	Briefly explain how the earned value analysis (EVA) technique is used in monitoring costs and progress in an IT project.	(4)
Q.4.3	In your own words, discuss what a baseline budget is and the relevance of developing one for any project.	(4)
Q.4.4	Using examples, explain how a project manager can effectively control project costs.	(6)

Paper 5: LU9

Question 6		(Marks: 20)
Answer all the questions. Acquiring a project team is one of the most important roles and responsibilities for a Project Manager.		
Q.6.1	List and discuss the four stages involved in acquiring a team in a project.	(12)
Q.6.2	Identify at least four barriers to team effectiveness in a project.	(4)
Q.6.3	What role does the Project Manager play in relation to the ethical behaviour of the team? What steps can be taken to help to ensure a high level of ethical behaviour?	(4)

Paper 6: LU9

Question 6		(Marks: 20)
Q.6.1	Describe any five (5) skills a project manager should have.	(5)
Q.6.2	Using suitable examples, explain any three (3) project team development stages.	(9)
Q.6.3	List two (2) advantages of each of the following project management organisational structures: • Functional structure; • Autonomous project structure.	(6)